

Three Sum

Problem ID: threesum

Given an array of numbers and a number k , determine if there are three entries in the array which add up to the specified number k .

For example, given $[20, 303, 3, 4, 25]$ and $k = 49$, return `true` as $20 + 4 + 25 = 49$.

Input

The input consists of two lines:

- The first line contains two integers n and k , where n is the number of elements in the array.
- The second line contains n space-separated integers representing the array.

Constraints:

- $3 \leq n \leq 2000$
- Array values fit in 32-bit signed integers
- k fits in a 32-bit signed integer

Output

Output `true` if there exist three numbers in the array whose sum is exactly k . Output `false` otherwise.

Sample

Sample Input 1	Sample Output 1
5 10 1 2 3 4 5	True